Sreekumar Kallingapuram Suresh

Thrissur, India | sreeku1993@gmail.com | +91 9567197101

Professional Summary

AI Engineer with an MSc in Artificial Intelligence and 4+ years of software engineering and QA automation experience. Specialized in building LLM-powered applications, RAG pipelines, and multimodal AI systems using Python, PyTorch, LangChain, and Google Gemini.

Experienced in designing and deploying AI solutions including career recommendation systems, medical document assistants, and semantic search engines. Strong background in scalable backend systems, automation, and AI application development.

Skills

Programming & Scripting

Python, JavaScript, TypeScript, Java, C++, HTML, CSS

AI / Machine Learning & GenAI

LLM Applications (Google Gemini, LLaMA), Retrieval-Augmented Generation (RAG), LangChain, FAISS Vector Search, Prompt Engineering, Multimodal AI Systems, Computer Vision (OpenCV), YOLO, OCR (EasyOCR), PDF Processing (pdfreader, pdfplumber), PyTorch, TensorFlow, Scikit-learn, SentenceTransformers(MiniLM)

AI Application Development

Next.js, Streamlit, Flask APIs, Supabase, REST APIs, Hugging Face Spaces, Vector-based Retrieval Systems, FastAPI

Cloud & DevOps

Google Cloud Platform (Cloud Run), Docker, Kubernetes, Git, CI/CD (Jenkins), Vercel (Next.js deployment)

Databases & Data Processing

MySQL, SQL Server, NoSQL (basic), Supabase, Pandas, Data preprocessing

Automation & Testing

Selenium WebDriver, Cypress, Playwright, TestNG, Postman, UFT

Tools & Platforms

VS Code, Jira, Confluence, Xray, Google App Scripts, Android Studio

Security & AI Safety

Hallucination Reduction Strategies (RAG validation), Privacy-aware Prompting, LLM Output Safety Filtering(Guardrails)

Development Methodologies

Agile / Scrum, Test Strategy, QA Automation

Selected AI & Machine Learning Projects

AI Career Recommendation Platform (LLM + Next.js)

- Designing and developing an **AI-powered SaaS platform** that analyzes user profiles and resumes to generate personalized career transition recommendations.
- Implemented **LLM-powered** analysis using **Google Gemini** to produce career summaries, role match scores, skill gap analysis, and structured 30/60/90 learning roadmaps.
- Designed a **context-augmented LLM pipeline** that collects structured career data and combines it with parsed resume content for deeper profile analysis.
- Developed the full-stack application using **Next.js, Supabase, and serverless deployment on Vercel**.
- Designing a scalable architecture for **AI-powered career coaching with subscription plans, payment integration, and interactive dashboard chat assistant**.
- Currently expanding the platform with **vector search, contextual chat assistants, and personalized learning recommendations**.

Assessment Recommendation System (NLP + Vector Search)

- Built a semantic recommendation engine to match job descriptions with relevant SHL assessments using **SentenceTransformers (MiniLM) embeddings and FAISS vector search**
- Implemented a **FastAPI backend** with ranking and filtering logic (duration, remote support, adaptive assessments) and evaluated performance using **Recall@10**
- Generated embeddings, indexed assessments with **FAISS**, and improved ranking using **hybrid similarity scoring with keyword boosting**
- Deployed the API on **Google Cloud Run** and developed a **Streamlit interface** for interactive testing of recommendations

---

AI Medical Assistant (Gemini, OCR, Python, Streamlit)

- AI system that analyzes **blood reports, X-rays, ECGs**, and medical documents
- Extracts structured data using **EasyOCR, pdfplumber, OpenCV**

- Generates **safe, patient-friendly medical summaries** using multimodal LLMs (Gemini/Llama)
- Designed **clinical-style reasoning prompts** (Findings → Severity → Recommendations)
- Focused on **health-tech domain**, privacy, and model safety

---

YOLO Object Detection – Nuts & Bolts Counter

- Applied **YOLO (CNN-based)** deep learning model for defect/part detection
- Processes video input and counts objects in industrial conveyor flow
- Improved detection speed and accuracy using custom dataset training

---

Professional Experience

Team Leader (Technical)

Planthood – London, UK | May 2024 – August 2025

- Designed, developed, and continuously improved an Android-based KPI tracking application integrated with Google Sheets for real-time monitoring.
- Built automation dashboards using Google Cloud and Apps Script to improve production analytics.
- Managed ERP systems and labeling infrastructure supporting daily production workflows..
- Coordinated cross-functional teams to optimize operational efficiency and reduce manual processes.

Programmer Analyst

Cognizant – Kochi, India | May 2021 – Dec 2021

- Collaborated with developers and analysts to design test strategies and validate system functionality.
- Performed API testing using Postman and data validation using SQL queries on Google Cloud databases.
- Led testing activities for an AI-based voice biometrics system and E-commerce platform for a major US telecom provider.
- Contributed to automation testing of E-commerce platform.

Test Engineer

Infosys Ltd. – Mysore, India | Aug 2018 – May 2021

- Developed and maintained automation frameworks using Selenium, UFT.
- Implemented CI/CD pipelines to automate testing and deployment workflows.
- Conducted end-to-end testing of e-commerce platforms including checkout flows and payment gateways.

Education

MSc in Artificial Intelligence – University of East London | Jan 2022 – May 2023

Master of Computer Applications – Anna University | Aug 2015 – Jun 2018

Bachelor of Computer Applications – Christ College | Aug 2011 – Aug 2014